Federated Learning and EHDS as Policy Data Infrastructure for Territorial Health Planning in Ageing Europe

Fabio Liberti
Department of Computer Science
Universitas Mercatorum
Rome, Italy
fabio.liberti@studenti.unimercatorum.it
ORCID: 0000-0003-3019-5411

***Abstract*—Demographic decline, population ageing and the progressive desertification of healthcare services in many European territories pose major challenges for territorial cohesion and equitable access to care. The European Health Data Space (EHDS) establishes a new regulatory framework for the secondary use of health data across Member States, opening opportunities for advanced analytics under strict data protection safeguards. This paper argues that federated learning (FL), when embedded within the EHDS governance framework, can serve as a *policy data infrastructure* for territorial healthcare planning in ageing and shrinking regions. Rather than focusing on algorithmic performance, we develop a conceptual model linking FL-enabled distributed analytics to policy questions on long-term care demand, service availability and territorial inequalities. We map policy-makers' information needs, analyse how EHDS provisions can underpin multi-territorial FL projects, and propose a framework describing actors, processes and outputs of EHDS-compliant FL initiatives for health and social care planning.**

***Index Terms*—Federated Learning, European Health Data Space, Territorial Health Planning, Demographic Change, Privacy-Enhancing Technologies, Health Data Governance**

I. Introduction

Demographic decline and rapid population ageing are reshaping the territorial distribution of health and social care needs across Europe. Shrinking and ageing regions, especially rural and inner areas, face increasing burdens of chronic and multi-morbid patients while experiencing a progressive "desertification" of healthcare and long-term care services. These dynamics pose major challenges for territorial cohesion and for ensuring equitable access to essential services, directly implicating the UN Sustainable Development Goals, particularly SDG 3 (Good Health and Well-Being) and SDG 10 (Reduced Inequalities).

Yet policy-makers at European, national and regional levels still rely on fragmented, siloed and poorly interoperable data, often confined within institutional or regional boundaries [12], [14]. This hampers their ability to anticipate future needs, identify territories at risk of medical desertification, and evaluate the territorial impact of reforms. Addressing demographic change and spatial inequalities in access to care requires granular, longitudinal, multi-source data and analytical capabilities that cut across organisational and jurisdictional borders [11], [13].

The European Health Data Space (EHDS) Regulation introduces a new regulatory and infrastructural framework for the primary and secondary use of electronic health data across Member States, with the explicit aim of fostering research, innovation and evidence-informed health policy. In parallel, federated learning (FL) has emerged as a privacy-enhancing paradigm for distributed analytics, allowing health data to remain under the control of local data holders while enabling the training of multi-centre models. We argue that, when combined, EHDS and FL can be conceived as a *policy data infrastructure* for territorial health planning in ageing and shrinking regions.

Rather than focusing on algorithmic performance or system implementation, this paper adopts a socio-technical and policy-oriented perspective. Our contributions are threefold: (1) We map key information needs of policy-makers dealing with demographic change, territorial cohesion and inequalities in access to care. (2) We analyse how EHDS provisions on secondary use, Health Data Access Bodies (HDABs) and privacy-enhancing technologies can underpin multi-territorial FL projects addressing these needs. (3) We propose a conceptual framework that describes actors, processes and outputs of EHDS-compliant FL initiatives, positioning FL as a policy data infrastructure for health and social care planning in demographically challenged territories.

The remainder of the paper is organised as follows. Section II reviews policy challenges at the intersection of demographic change, territorial cohesion and access to care. Section III surveys related work on FL in healthcare, FL governance and EHDS. Section IV introduces the EHDS regulatory framework for the secondary use of health data. Section V conceptualises FL as a policy data infrastructure and maps typical policy questions to EHDS data and FL-enabled outputs. Section VI presents a policy-oriented framework for EHDS-compliant FL projects. Section VII discusses implications for territorial cohesion, sustainability of health systems and future

research.

II. Demographic Change, Territorial Cohesion and Access to Care

European territories are undergoing heterogeneous trajectories of demographic change [21]. The EU population is projected to decline by 3.8% by 2070, while the old-age dependency ratio is expected to rise from 33.4% to 56.7% [21]. While some metropolitan regions continue to grow, many rural, peripheral and inner areas experience sustained population decline and accelerated ageing. These trends translate into shrinking local tax bases, rising dependency ratios and difficulties in maintaining dense networks of health and social care services. In several Member States, this has given rise to "medical desertification", characterised by shortages of general practitioners, closure or downgrading of small hospitals, and long travel times to access basic services.

Territorial inequalities in access to care have thus become a central concern for both health and cohesion policies [11]. Residents in shrinking and ageing areas often face higher barriers to timely diagnosis and treatment, limited availability of long-term care options and greater reliance on informal care. At the same time, regional and local health authorities struggle to plan and finance services in territories with small, dispersed populations, where economies of scale are difficult to achieve and traditional hospital-centred models of care are not sustainable.

Against this backdrop, policy-makers face a set of recurring information needs. First, they need to anticipate the spatial distribution of future demand for chronic and long-term care under different demographic scenarios. Second, they must identify territories at high risk of medical desertification and avoidable inequalities in access to essential services. Third, they require tools to assess the potential impact of alternative configurations of services—such as the closure or consolidation of small hospitals, the development of community-based facilities, or the deployment of telemedicine—on access and outcomes across territories. Finally, they need to monitor over time the territorial effects of reforms and cohesion policy interventions [11], [21].

Meeting these information needs is challenging. Relevant data are typically scattered across multiple organisations (regional health authorities, hospitals, primary care organisations, social care providers, national registries), governed by different legal regimes and stored in heterogeneous formats. Cross-border and multi-regional analyses are further hampered by differences in coding practices and infrastructures. As a result, many decisions concerning the organisation of services in ageing and shrinking territories are taken with partial or delayed evidence, even when rich data exist locally but cannot be easily combined or analysed at scale [12], [13].

III. Related Work

Federated learning has attracted growing attention in healthcare as a way to enable multi-centre analysis of sensitive data while preserving institutional control and complying with data protection regulations [4]–[6]. Systematic reviews document a rapid expansion of FL applications across domains such as oncology, radiology and chronic disease management, but also highlight technical and organisational challenges related to data heterogeneity, communication efficiency and evaluation [4], [5], [7], [23]. Yet only 5.2% of FL healthcare studies achieve real-life application [7], and sustained production deployment remains at 23%, with hardware heterogeneity and non-IID data as dominant barriers [17]. Notable exceptions such as the EXAM COVID-19 multi-centre study demonstrate that large-scale FL in healthcare is feasible when governance and technical conditions are met [18]. More recently, a stream of work has begun to address FL in broader public health and population-level contexts, for example in disease surveillance, outbreak detection and risk stratification for prevention-oriented policies [7], [25].

In parallel, scholars have started to examine the *governance* of FL in healthcare. Scoping reviews and conceptual frameworks emphasise that federated approaches require tailored governance mechanisms beyond those used for centralised machine learning, including new procedural, relational and structural arrangements [8], [9]. Proposed frameworks underline the need for clear roles and responsibilities, formal agreements, oversight structures and stakeholder engagement to address ethical concerns, fairness, accountability and the risk of harm [8], [10]. A recent Baltic-Nordic roadmap further integrates sustainability, regulatory alignment and capacity building across the FL project lifecycle [26]. However, empirical evidence on how such governance models operate in practice remains limited and fragmented [8].

On the regulatory side, the emerging literature on the European Health Data Space (EHDS) stresses its potential to overcome the current fragmentation of health data in Europe while balancing opportunities for data reuse with risks to privacy and fundamental rights [1]–[3], [19]. Authors note that the EHDS explicitly recognises the role of privacy-enhancing technologies (PETs), including federated systems, as tools to implement secure secondary use under the supervision of Health Data Access Bodies and within secure processing environments [3]. Implementation readiness, however, varies significantly across Member States [12], [14]. At the same time, interviews with stakeholders reveal techno-legal uncertainties and gaps in the adoption of PETs within EHDS-like frameworks, calling for further conceptual and empirical work on how these technologies can be embedded in governance structures [3], [15].

While these strands of research provide important foundations, they leave at least two gaps that this paper addresses. First, existing studies on FL in healthcare and public health seldom consider *demographic decline, territorial cohesion and inequalities in access to care* as primary lenses, focusing instead on clinical or disease-specific use cases [6], [7], [23]. Second, most discussions of EHDS and PETs remain at a high level, without articulating how federated learning could be operationalised as a *policy data infrastructure* for territorial health planning within the EHDS governance framework. To

the best of our knowledge, this paper is the first to bridge these two gaps by systematically conceptualising FL within the EHDS as a policy data infrastructure for territorial health planning in demographically challenged regions. A companion technical paper [20] demonstrates the feasibility of EHDS-compliant FL through a three-layer reference architecture and 5,477+ experiments showing that privacy at ε=10 costs less than 2 percentage points and full EHDS compliance is essentially free for personalised algorithms. Building on that technical foundation, this paper connects the governance, policy and territorial literatures to conceptualise EHDS-compliant FL projects that explicitly target the challenges of ageing and shrinking territories.

IV. The European Health Data Space as a Regulatory Infrastructure

The EHDS Regulation establishes a common framework for the use and exchange of electronic health data across the EU, with the dual objective of empowering individuals and enabling secure data reuse for research, innovation and policy-making [1], [2]. It seeks to overcome the current fragmentation of health data landscapes by harmonising legal and technical conditions for data access, interoperability and quality, while preserving strong protections for privacy and data protection. The EHDS operates as *lex specialis* alongside the General Data Protection Regulation (GDPR), adding sector-specific rules for health data access and reuse while maintaining full compatibility with the GDPR's fundamental safeguards [1], [2]. Furthermore, FL-based systems supporting healthcare decision-making may qualify as high-risk under the EU Artificial Intelligence Act (Regulation (EU) 2024/1689), adding transparency, data governance and human oversight requirements to the regulatory landscape.

For secondary use, the EHDS creates a secure, consistent and trustworthy framework that allows electronic health data to be reused for predefined purposes, including health-related scientific research, training and testing of algorithms, and policymaking in the healthcare sector [1], [2]. Member States must establish Health Data Access Bodies (HDABs), which serve as gatekeepers: they receive and assess requests from data users, issue data permits, and ensure that data processing takes place in secure processing environments [1], [2]. Data holders are required to catalogue in-scope datasets and make them available for secondary use under the conditions specified in data permits, typically in anonymised or pseudonymised form and without allowing raw data to be downloaded [1], [2].

From the viewpoint of territorial health planning, the EHDS is particularly relevant because it enables cross-organisation and cross-border analyses under a shared governance regime. HDABs and the HealthData@EU infrastructure are intended to support multi-country projects [16], while interoperability requirements—building on standards such as HL7 FHIR and the European Electronic Health Record Exchange Format (EEHRxF)—and common data categories facilitate the combination of data from different regions and providers [1], [2], [13]. However, the Regulation does not prescribe specific analytical architectures. It allows for different technical approaches, including centralised data warehouses, data enclaves and privacy-enhancing technologies such as federated learning, as long as they comply with the conditions set out in data permits and secure processing requirements [3]. At the same time, implementation readiness varies considerably: Nordic countries are 2–3 years ahead in HDAB capacity-building, while data access timelines range from 3 weeks to over 12 months across Member States [12], [14]. This heterogeneity shapes the conditions under which FL can be deployed, but the regulatory openness nonetheless creates space to explore FL as a distributed, policy-oriented way of exploiting EHDS-enabled data resources.

V. Federated Learning as a Policy Data Infrastructure

We use the term *policy data infrastructure* to denote the ensemble of regulatory frameworks, institutional arrangements, technical standards and analytical capabilities that enable the systematic production and use of data-driven evidence for public policy purposes. We argue that EHDS and FL jointly constitute such an infrastructure when oriented towards territorial health planning: EHDS provides the governance, legal and interoperability layers, while FL provides the distributed analytical engine that operates within those layers.

A. From Clinical to Population-Level Analytics

As reviewed in Section III, FL enables multi-institutional model training without centralising raw data, preserving data sovereignty and regulatory compliance [4], [5]. In this paper we focus on *horizontal cross-silo* FL, the most natural setting for territorial health planning: each data holder (e.g. a regional health authority or hospital) holds records with similar features for different patient populations, and models are trained collaboratively across these institutional silos. While most healthcare applications to date have focused on clinical prediction, the same principles can support population-level analytics for public health and resource allocation [7], [18], [25]. Recent experimental evidence demonstrates that FL can achieve performance within 0.5 percentage points of centralised approaches on adequately-sized European clinical datasets, with privacy guarantees at ε=10 costing less than 2 percentage points [20]. In this broader perspective, federated networks of health data holders can generate insights that inform policy decisions while respecting the distribution of responsibilities and control over data across institutions [4], [22].

B. Policy Functions Enabled by FL Within EHDS

When deployed within the governance framework of the EHDS, FL can act as a *policy data infrastructure* for territorial health planning. Rather than being used only at the point of care, FL can support analyses that address the information needs identified in Section II:

- **Forecasting demand**: FL can estimate current and future demand for chronic care and long-term care across territories by training models on distributed clinical and utilisation data without centralising records [23], [25].
- **Identifying territorial risks**: Multi-regional FL models can help identify areas at high risk of medical desertification or avoidable inequalities in access, combining information on service availability, utilisation patterns and population characteristics [7], [22].
- **Scenario analysis**: FL-enabled models can support simulation of the territorial impact of alternative service configurations (e.g. reorganising hospitals, expanding community care, deploying telemedicine), under the conditions set in EHDS data permits and secure environments [3], [20].

In all these cases, EHDS provides the legal and institutional layer (HDABs, data permits, secure environments, standardised data domains), while FL provides the analytical mechanism to exploit distributed data sources for policy purposes. The EHDS Regulation also recognises other privacy-enhancing technologies, such as secure multi-party computation (SMPC) and trusted execution environments (TEEs) [3]. FL is well suited for territorial planning because it scales to many heterogeneous data holders, produces reusable trained models rather than one-off query results, and can be combined with differential privacy to provide formal guarantees [24]—properties that are harder to achieve jointly with SMPC and TEEs at the scale of multi-regional health systems [3].

C. Mapping Policy Questions to EHDS Data and FL-Enabled Outputs

Table I offers a conceptual mapping between typical policy questions arising in demographically challenged territories, the EHDS data categories that may be involved, the role of FL, and the types of outputs that can inform decision-making. This mapping is illustrative rather than exhaustive. Its purpose is to show how EHDS-enabled data, combined through FL rather than centralisation, can produce outputs that are directly actionable for territorial health and social care planning.

D. Visual Synthesis of the EHDS-FL Synergy

Fig. 1 provides a composite visual synthesis of the proposed policy data infrastructure. Panel (a) presents the end-to-end architecture: from the societal context (demographic decline, ageing, medical desertification) through policy-maker needs, the EHDS regulatory framework and the FL analytical engine, to the project framework and illustrative scenario examples. Panel (b) details the operational framework at three levels: (A) the strategic context linking territorial challenges to policy solutions through EHDS and FL; (B) the EHDS-compliant actor ecosystem with roles and interactions among policy sponsors, HDABs, analytical coordinators and data holders/FL participants; and (C) the five-stage project lifecycle from problem formulation through policy uptake and feedback.

In the next section, we elaborate on the actors, processes and outputs illustrated in Fig. 1 by proposing a detailed policy-oriented framework for implementing EHDS-compliant FL projects within demographically challenged territories.

VI. A Policy-Oriented Framework for EHDS-Compliant FL Projects

This section proposes a conceptual framework for designing and governing federated learning projects for territorial health planning within the EHDS. The framework focuses on actors, processes and outputs rather than on technical architectures.

A. Actors and Roles

Building on the EHDS governance structures and as illustrated in Fig. 1b(B), we identify four main groups of actors involved in EHDS-compliant FL projects for demographically challenged territories:

- **Policy sponsors** (e.g. ministries of health, regional health authorities, cohesion policy bodies) define the policy questions, prioritise territories and provide strategic direction and resources.
- **Health Data Access Bodies (HDABs)** act as gatekeepers for secondary use: they receive project proposals, assess legal and ethical compliance, issue data permits and monitor adherence to conditions, including the use of privacy-enhancing technologies [1], [2].
- **Data holders and FL participants** (e.g. regional health authorities, hospitals, primary care organisations, social care agencies) control relevant datasets and host local FL components, preparing data in accordance with EHDS interoperability requirements and project specifications.
- **Analytical coordinators** (e.g. public health institutes, research organisations, competence centres) design the FL studies, coordinate training rounds across participants, and synthesise outputs into policy-oriented insights such as indicators, maps and scenarios.

Effective projects require clear allocation of responsibilities and mutual trust among these actors [9], [10], as well as adequate technical and organisational capacity on the side of data holders and coordinators.

B. Policy and Data Lifecycle

We conceptualise the lifecycle of an EHDS-compliant FL project for territorial planning in five stages, as depicted in Fig. 1b(C):

1. Problem formulation. Policy sponsors articulate a territorial challenge (e.g. ensuring access to long-term care in shrinking inner areas) and specify high-level information needs, time horizons and target populations. This stage requires dialogue between policy and analytical communities to translate policy questions into analysable problems.

2. Project design and authorisation. Analytical coordinators, in consultation with potential data holders, design a FL-based project that identifies relevant EHDS data categories, participating organisations, privacy-enhancing techniques, expected outputs and dissemination plans. A proposal is submitted to the HDAB, which evaluates legal bases, safeguards,

TABLE I: Federated learning within EHDS as a policy data infrastructure for territorial planning.

Policy question	EHDS data categories	EHDS-specific enabler	Role of FL	Policy-relevant outputs
How will LTC demand evolve in shrinking, ageing areas?	EHRs, chronic diagnoses, admissions, prescriptions, functional status	HDAB data permits; secure processing environments	Train distributed risk and utilisation models across regions	Territorial risk maps and LTC demand projections
Which territories face medical desertification and poor access?	Provider registries, utilisation, waiting times, demographic indicators	HealthData@EU cross-border infrastructure; standardised data categories	Identify patterns of under-/over-provision across territories	Territorial profiles and rankings for investment prioritisation
What is the impact of reorganising hospitals and community services?	Historical utilisation, outcomes, mobility flows, demographic trends	Multi-HDAB permit coordination; interoperability standards	Scenario-based modelling of service configurations	Comparative scenarios for planners, highlighting territorial winners and losers

proportionality and alignment with EHDS purposes before issuing a data permit with specific conditions [1]–[3].

3. Distributed analysis. Under the permit, participating data holders prepare their data according to agreed formats and standards and deploy local FL components. Training rounds are executed in secure environments, without transferring raw data, and respecting constraints on data access and logging imposed by the permit and the HDAB.

4. Synthesis and interpretation. Analytical coordinators aggregate FL outputs, compute indicators and generate territorial maps and scenarios aligned with the original policy questions. This stage involves interdisciplinary interpretation, including public health, economics and territorial planning perspectives, and may require simplification and visualisation to support decision-makers.

5. Policy uptake and feedback. Results inform planning documents, investment decisions and reform options (e.g. location of new community services, reconfiguration of hospital networks, targeting of cohesion funds). Implementation feedback and new data may trigger further analytical cycles, enabling iterative refinement of both models and policies.

This lifecycle is consistent with emerging roadmaps for sustainable FL in Europe [26] and emphasises that FL is embedded within institutional processes defined by EHDS, rather than being an isolated technical solution. It also highlights the importance of aligning analytical design with governance requirements and policy timetables: data permits have temporal bounds and purpose limitations that constrain the scope of FL projects, while policy cycles impose deadlines that the analytical workflow must respect.

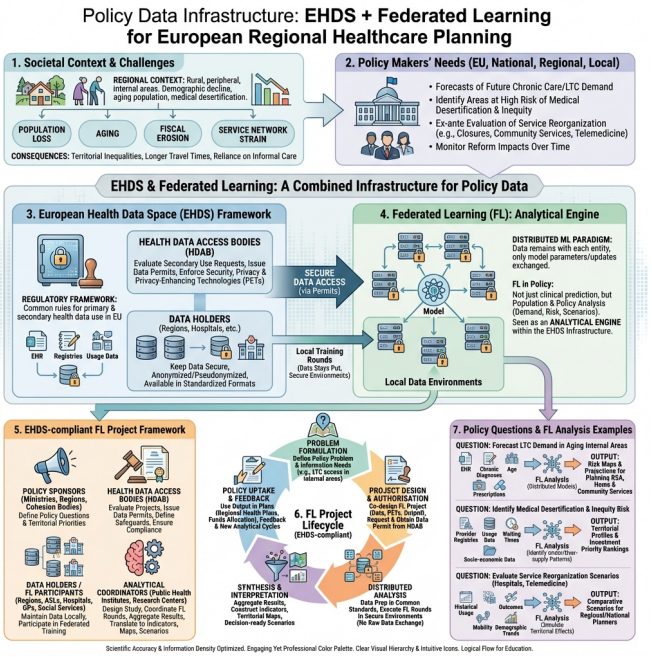
C. Implementation Challenges

Operationalising the proposed framework raises several challenges that deserve explicit attention. First, *data quality and harmonisation* across participating institutions remain a major obstacle: even within the EHDS interoperability framework, differences in coding practices, data completeness and update frequency can introduce significant heterogeneity into FL training processes [13]. Second, *capacity asymmetries* among data holders may limit participation: smaller hospitals or social care agencies in peripheral territories may lack the technical infrastructure, computational resources or trained personnel needed to deploy FL components [12], [17]. Third, *governance complexity* increases with the number of jurisdictions involved: cross-border projects require coordination among multiple HDABs, each operating under partially different national implementations of the EHDS Regulation [14], [15]. Fourth, *stakeholder trust* must be actively built: citizens, clinicians and local administrators need to understand and accept that FL projects process health data for policy purposes, even when raw data remain local [19]. Fifth, *algorithmic fairness across territories* requires attention: if data from well-resourced, digitally mature regions are richer or more complete, federated models may yield less accurate predictions for the very peripheral territories the framework aims to serve, risking the paradox of an equity-oriented infrastructure that amplifies existing disparities [20], [24]. Addressing these challenges requires not only technical solutions but also investment in capacity building, institutional cooperation, public engagement and fairness-aware model design.

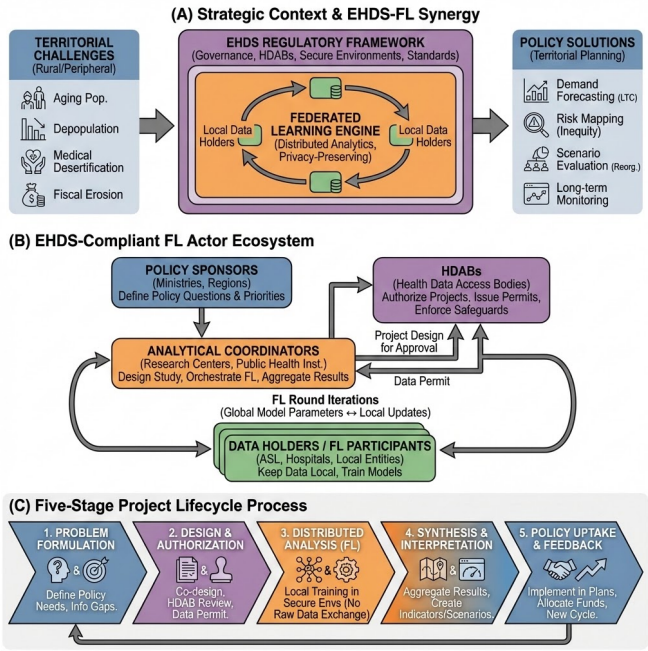
D. From FL Outputs to Territorial Decisions

The value of EHDS-compliant FL projects ultimately depends on their ability to influence territorial decision-making. We identify three key mechanisms through which FL outputs can be integrated into policy processes:

- **Evidence for strategic planning.** FL-based projections of long-term care demand, risk maps of medical desertification and comparative territorial profiles can be incorporated into regional and national health plans, as well as into programmes for inner areas and cohesion policy.
- **Scenario-informed deliberation.** Scenario analyses that compare alternative configurations of services provide a



(a) End-to-end policy data infrastructure: EHDS + Federated Learning for European regional healthcare planning.



(b) EHDS-FL synergy: strategic context, actor ecosystem and project lifecycle.

Fig. 1: Composite framework for EHDS-compliant federated learning as policy data infrastructure. (a) End-to-end view showing six interconnected blocks: societal context and challenges, policy-maker information needs, the EHDS regulatory framework for secondary use, the FL analytical engine, the EHDS-compliant project framework, and illustrative policy scenarios with expected outputs. (b) Operational detail at three levels: (A) strategic context linking territorial challenges to policy solutions; (B) actor ecosystem with roles and interactions among policy sponsors, HDABs, analytical coordinators and data holders/FL participants; (C) five-stage project lifecycle from problem formulation to policy uptake and iterative feedback.

structured basis for deliberation among policy-makers, clinicians, local authorities and citizens, helping to make explicit trade-offs between access, efficiency and sustainability.

- **Accountability and monitoring.** Repeated FL analyses over time can support monitoring of territorial impacts of reforms and targeted interventions, providing a basis for accountability and for adjusting strategies in light of observed outcomes [26].

To fulfil these functions, FL outputs must be interpretable, transparent and sensitive to issues of fairness across territories. This implies that FL projects for territorial planning should not only optimise predictive performance, but also explicitly consider how results will be communicated and used in decision-making arenas.

VII. Discussion and Conclusions

A. Implications for Research and Practice

Conceptualising federated learning within the EHDS as a policy data infrastructure for territorial health planning has several implications for research and practice. First, it shifts attention from purely technical considerations towards the institutional, organisational and governance arrangements required to make distributed analytics useful for addressing demographic decline, ageing and territorial inequalities in access to care. In doing so, it aligns the debate on federated learning with broader agendas on territorial cohesion and sustainability of health and social care systems.

Second, the framework provides a structured vocabulary for dialogue between data scientists, health planners and governance bodies. By identifying four actor groups, five lifecycle stages and three policy integration mechanisms, it offers a reference architecture that can guide the design of concrete EHDS-compliant FL initiatives. This is particularly relevant as Member States approach the 2029 deadline for secondary use provisions and begin to operationalise HDABs and secure processing environments [12], [15].

Third, the conceptual mapping between policy questions, EHDS data categories and FL-enabled outputs (Table I) provides an actionable template that policy-makers can adapt to their specific territorial contexts. Rather than requiring deep technical expertise, the mapping allows decision-makers to articulate their information needs in terms that can be translated into FL project specifications by analytical coordinators.

B. Opportunities and Risks

The framework highlights both opportunities and risks. On the opportunity side, EHDS-compliant FL projects could

enable richer, more timely and more geographically granular evidence for planning services in shrinking and ageing territories, while respecting data protection and institutional autonomy. Experimental evidence confirms that full EHDS compliance—simultaneous data minimisation, citizen opt-out and differential privacy—costs personalised FL algorithms as little as 0.7 percentage points [20], making the privacy–utility trade-off highly favourable for policy applications. Moreover, FL can bridge data silos that have historically prevented comprehensive territorial analyses: by federating models across regional health authorities, hospitals and social care agencies, it enables a holistic view of care demand and provision that no single institution could achieve alone.

On the risk side, the complexity of EHDS governance and FL infrastructures may privilege digitally advanced regions and well-resourced organisations, potentially reinforcing existing inequalities if not carefully managed [12], [17]. This "digital divide" risk is particularly acute in the very territories that the framework aims to serve: shrinking, ageing and peripheral areas are often those with the least developed digital health infrastructure. Implementation readiness varies significantly across Member States [12], and only a minority of FL healthcare studies achieve real-world deployment [7], [17]. Ensuring meaningful participation of less-resourced territories, transparent communication about methods and limitations, and appropriate involvement of citizens and local stakeholders will be critical [19].

C. Limitations

This paper has several limitations. First, the framework is conceptual and has not been empirically validated through pilot implementations. While the companion technical paper [20] demonstrates the technical feasibility of EHDS-compliant FL, the institutional and organisational dimensions of the framework—multi-actor coordination, HDAB permit processes, policy uptake—require empirical testing in real governance contexts. Second, the paper focuses on the European context and the EHDS specifically; transferability to other regulatory environments (e.g. the US, Asia-Pacific) would require adaptation. Third, the policy questions and data mappings presented in Table I are illustrative; actual implementations would need to engage with the specific data availability, quality constraints and institutional arrangements of target territories. Fourth, the four-actor, five-stage framework is a deliberate simplification: in practice, actors may assume multiple roles, stages may overlap or iterate, and the relative weight of implementation challenges varies across territorial contexts. The framework does not prescribe priorities among these challenges, leaving such contextualisation to future empirical work.

D. Future Work

The analysis points to an agenda for future work along three lines. First, empirical studies are needed to assess the feasibility and added value of EHDS-compliant FL projects in real policy contexts, including pilot projects in selected regions and cross-country comparisons. Learnings from the HealthData@EU Pilot [16] and from ongoing HDAB implementation efforts [15] should inform the design of such initiatives. Second, research should explore how different governance models for Health Data Access Bodies and data spaces affect the design and uptake of FL-based policy analytics [8], [10]. Comparative studies across Member States with different levels of digital health maturity would be particularly valuable. Third, there is a need for interdisciplinary approaches that integrate data science, health policy, territorial planning and ethics to ensure that FL-enabled infrastructures contribute to fair and sustainable responses to demographic change. This includes developing methods for assessing diagnostic equity across territories, building on recent work on fairness metrics in federated settings [20].

E. Concluding Remarks

While this paper has deliberately taken a conceptual and framework-oriented stance, it aims to provide a foundation for empirical and design-oriented endeavours. By framing FL as a policy data infrastructure embedded in the EHDS, it provides a structured basis for investigating how advanced analytics can be responsibly mobilised to support territorial cohesion and equitable access to care in an ageing Europe. The convergence of a mature regulatory framework (EHDS), a technically viable analytical paradigm (FL) and an urgent societal challenge (demographic decline) creates a unique window of opportunity for evidence-informed territorial health planning that this paper has sought to articulate and structure. In doing so, it reinforces the contribution to SDG 3 and SDG 10 outlined in the Introduction, by advancing a data infrastructure that can support equitable access to care across demographically challenged territories.

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References

[1] European Parliament and Council, "Regulation (EU) 2025/327 of 11 February 2025 on the European Health Data Space and amending Directive 2011/24/EU and Regulation (EU) 2024/2847," *Official Journal of the European Union*, 2025.

[2] European Commission, "European Health Data Space Regulation (EHDS) – Public Health," 2025. [Online]. Available: https://health.ec.europa.eu/ehealth-digital-health-and-care/european-health-data-space-regulation-ehds_en

[3] S. van Drumpt, P. De Hert *et al.*, "Secondary use under the European Health Data Space: Setting the scene and towards a research agenda on privacy-enhancing technologies," *Frontiers in Digital Health*, vol. 7, art. 1602101, 2025.

[4] N. Rieke *et al.*, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, art. 119, 2020.

[5] B. Pfitzner, N. Steckhan, and B. Arnrich, "Federated learning in a medical context: A systematic literature review," *ACM Trans. Internet Technol.*, vol. 21, no. 2, pp. 1–31, 2021.

[6] M. G. Crowson *et al.*, "A systematic review of federated learning applications for biomedical data," *PLOS Digital Health*, vol. 1, no. 5, art. e0000033, 2022.

[7] Z. L. Teo *et al.*, "Federated machine learning in healthcare: A systematic review on clinical applications and technical architecture," *Cell Reports Medicine*, vol. 5, no. 2, art. 101419, 2024.

[8] R. Eden *et al.*, "A scoping review of the governance of federated learning in healthcare," *npj Digital Medicine*, vol. 8, art. 427, 2025.
[9] T. Barbereau, J. Delgado Fernandez, and S. Potenciano Menci, "The governance of federated learning: A decision framework for organisational archetypes," *Data & Policy*, vol. 7, art. e53, 2025.
[10] J. van Soest *et al.*, "A governance framework for federated learning projects in healthcare," *Studies in Health Technology and Informatics*, vol. 327, pp. 258–262, 2025.
[11] WHO and UN-Habitat, *Health at the Heart of Urban and Territorial Planning*. Geneva: WHO, 2021.
[12] TEHDAS Joint Action, "Are EU member states ready for the European Health Data Space?" *Eur. J. Public Health*, vol. 34, no. 6, pp. 1102–1108, 2024.
[13] R. Hussein *et al.*, "Interoperability framework of the EHDS for secondary use," *J. Med. Internet Res.*, vol. 27, art. e69813, 2025.
[14] R. Forster *et al.*, "User journeys in cross-European secondary use of health data," *Eur. J. Public Health*, vol. 35, Suppl. 3, pp. iii18–iii24, 2025.
[15] L. Svingel *et al.*, "Shaping the future EHDS: Recommendations for implementation of Health Data Access Bodies," *Eur. J. Public Health*, vol. 35, Suppl. 3, pp. iii32–iii38, 2025.
[16] C. Christiansen *et al.*, "Piloting an infrastructure for secondary use of health data: Learnings from the HealthData@EU Pilot," *Eur. J. Public Health*, vol. 35, Suppl. 3, pp. iii3–iii4, 2025.
[17] H. Fröhlich *et al.*, "Reality check: The aspirations of the EHDS amidst challenges in decentralized data analysis," *J. Med. Internet Res.*, vol. 27, art. e76491, 2025.
[18] I. Dayan *et al.*, "Federated learning for predicting clinical outcomes in patients with COVID-19," *Nature Medicine*, vol. 27, no. 10, pp. 1735–1743, 2021.
[19] C. Staunton *et al.*, "Ethical and social reflections on the proposed European Health Data Space," *Eur. J. Human Genetics*, vol. 32, no. 5, pp. 498–505, 2024.
[20] F. Liberti, "FL-EHDS: A privacy-preserving multimodal federated learning framework for the European Health Data Space," in *Proc. FLICS*, 2026 (to appear).
[21] European Commission, "The 2024 Ageing Report: Economic and Budgetary Projections for the EU Member States (2022–2070)," *Institutional Paper* 279, Luxembourg: Publications Office of the EU, 2024.
[22] M. J. Sheller *et al.*, "Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, art. 12598, 2020.
[23] Prayitno, C.-R. Shyu, K. T. Putra, H.-C. Chen, Y.-Y. Tsai, K. S. M. T. Hossain, W. Jiang, and Z.-Y. Shae, "A systematic review of federated learning in the healthcare area: From the perspective of data properties and applications," *Applied Sciences*, vol. 11, no. 23, art. 11191, 2021.
[24] M. Adnan, S. Kalra, J. C. Cresswell, G. W. Taylor, and H. R. Tizhoosh, "Federated learning and differential privacy for medical image analysis," *Scientific Reports*, vol. 12, art. 1953, 2022.
[25] T. S. Brisimi, R. Chen, T. Mela, A. Olshevsky, I. C. Paschalidis, and W. Shi, "Federated learning of predictive models from federated electronic health records," *Int. J. Med. Inform.*, vol. 112, pp. 59–67, 2018.
[26] J. Kommusaar, S. Elunurm, T. Chomutare, M. Kangasniemi, S. Salanterä, and L.-M. Peltonen, "A roadmap for federated learning projects using health data to guide sustainable artificial intelligence development in the European Union," *Int. J. Med. Inform.*, vol. 208, art. 106242, 2026.